

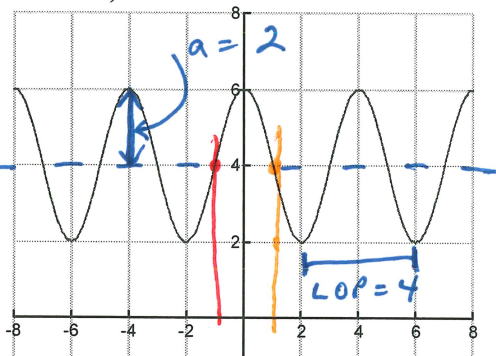
As discussed in class, many solutions are possible, but the a , b , and d values are always the same. The Lb and c values can change based on the type of equation and where you choose to put the left boundary.

AP Precalculus Unit 2 Wkst D

Name: _____

For each of the following write an equation of the indicated type that will generate the given graph.

1. a) Negative Sine
b) Positive Sine

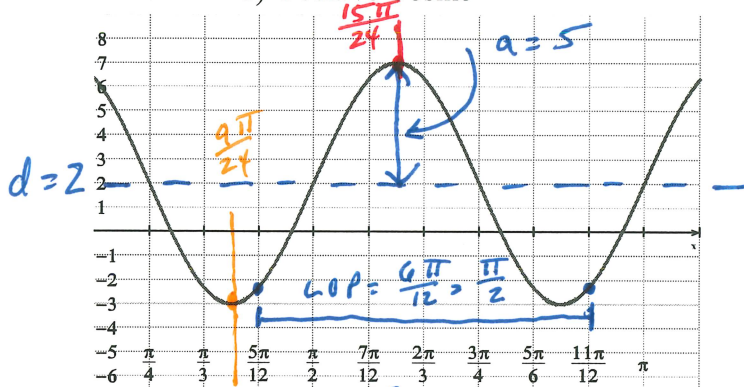


$$LOP = \frac{2\pi}{b} \quad b = \frac{2\pi}{4} = \frac{\pi}{2}$$

a) $LB = 1 \quad c = -1 \quad y = -2\sin\left(\frac{\pi}{2}(x-1)\right) + 4$

b) $LB = -1 \quad c = +1 \quad y = 2\sin\left(\frac{\pi}{2}(x+1)\right) + 4$

2. a) Negative Cosine
b) Positive Cosine

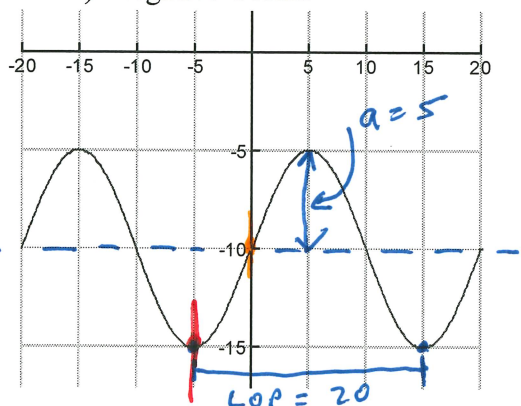


$$LOP = \frac{2\pi}{b} \quad b = \frac{2\pi}{\frac{\pi}{2}} = 4$$

a) $LB = \frac{9\pi}{24} \quad c = -\frac{9\pi}{24} \quad y = -5\cos\left(4\left(x - \frac{9\pi}{24}\right)\right) + 2$

b) $LB = \frac{15\pi}{24} \quad c = -\frac{15\pi}{24} \quad y = 5\cos\left(4\left(x - \frac{15\pi}{24}\right)\right) + 2$

3. a) Positive Sine
b) Negative Cosine

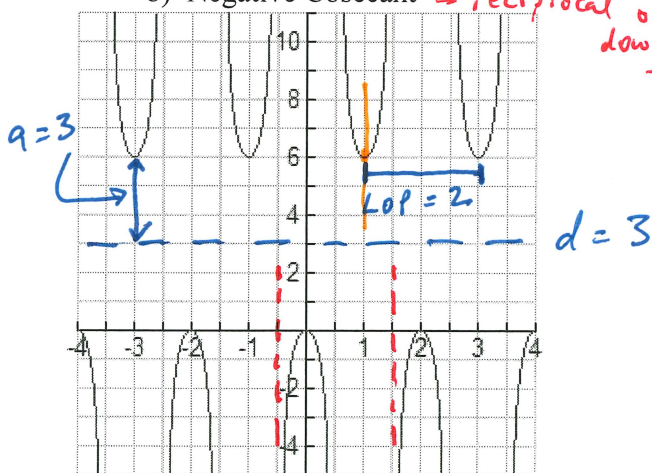


$$LOP = \frac{2\pi}{b} \quad b = \frac{2\pi}{LOP} = \frac{2\pi}{20} = \frac{\pi}{10}$$

a) $LB = 0 \quad c = 0 \quad y = 5\sin\left(\frac{\pi}{10}x\right) - 10$

b) $LB = -5 \quad c = +5 \quad y = -5\cos\left(\frac{\pi}{10}(x+5)\right) - 10$

4. a) Positive Secant — reciprocal of cosine up first
b) Negative Cosecant — reciprocal of sine down first



$$LOP = \frac{2\pi}{b} \quad b = \frac{2\pi}{2} = \pi$$

a) $LB = 1 \quad c = -1$

$$y = 3\sec\left(\pi(x-1)\right) + 3$$

b) $LB = -\frac{1}{2} \quad c = +\frac{1}{2}$

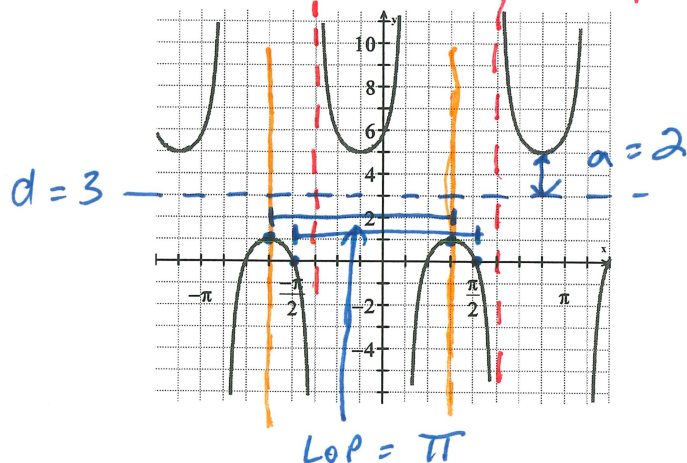
$$y = 3\csc\left(\pi\left(x + \frac{1}{2}\right)\right) + 3$$

5.

a) Negative Secant

b) Positive Cosecant

— reciprocal of cosine, down first
— reciprocal of sine, up first



$$Lop = \frac{2\pi}{b} \quad b = \frac{2\pi}{\pi} = 2$$

a) $LB = -\frac{5\pi}{8}$ $c = +\frac{5\pi}{8}$

$$y = -2 \sec\left(2\left(x + \frac{5\pi}{8}\right)\right) + 3$$

b) $LB = -\frac{3\pi}{8}$ $c = +\frac{3\pi}{8}$

$$y = 2 \csc\left(2\left(x + \frac{3\pi}{8}\right)\right) + 3$$